PREM P S TOMAR

Software Architect | Principal Engineer

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PROFESSIONAL SUMMARY

Software architect and hands-on senior individual contributor with 14+ years designing, modernizing and shipping distributed platforms. Combines Rust, C++, C and Python with AI/ML and Generative AI integration, complemented by Go and Java backend architecture and Node.js delivery. Defines service boundaries, integration contracts and cloud architecture while staying hands-on through implementation, performance investigation and production reliability. Builds reusable platforms across fintech, insurance, healthcare, media, telecom, commerce and logistics; mentors engineers and balances scale, security, cost and release velocity.

CORE SKILLS

Systems programming: Rust, C++, C, Python; CPU and memory profiling, concurrent processing and high-volume data pipelines.

AI/ML and Generative AI: GenAI, AWS Bedrock, retrieval-augmented generation (RAG), LLM integration, multi-LLM gateways, ML decisioning, fraud analytics, sentiment signals, recommendations, probabilistic and forensic modeling.

Backend and architecture: Go (Golang), Java/Spring Boot, Node.js, TypeScript, FastAPI; distributed systems, microservices, REST APIs, gRPC-style contracts, API gateways, service decomposition, event-driven design.

Workflow and decisioning: BPMN, Camunda, Temporal, Flowable concepts, state machines, routing, approvals, SLA visibility, audit trails, case lifecycles and document-processing automation.

Data and event platforms: PostgreSQL/Aurora, MongoDB, Redis/Valkey, Kafka-style streams, Spark-style processing, Elasticsearch-style search, schema governance, data integrity, analytics and feature-ready pipelines.

Cloud and reliability: AWS, Kubernetes, Docker, OpenShift-aligned patterns, service mesh concepts, CI/CD, Git, monitoring, tracing, logging, profiling, production RCA and capacity planning.

AI-assisted engineering: AI tools and coding agents for bounded diff-set generation, refactoring analysis, test scaffolding and performance investigation; review each step, validate with tests and profiling, retain design ownership and code integrity.

Technical leadership: Architecture and code reviews, platform standards, API governance, roadmap and release planning, executive communication, cross-functional decisions, mentoring senior and architect-level engineers.

Security and compliance: PCI/SOC2-aligned workflows, secure service boundaries, privacy-aware data handling, auditability, regulated delivery and operational readiness.

PROFESSIONAL EXPERIENCE

DDN

Incoming Staff Engineer | Starts 2 November 2026

SwissClear Services Private Limited (TAPPP)

Senior Architect / Principal Solutions Architect | Feb 2024 - Oct 2026

- Directed architecture for distributed backend, runtime, workflow and micro-app platforms serving 7M+ daily users across media, sports, fintech and engagement products.
- Consolidated client-specific capabilities into reusable modules, shared service contracts and configurable onboarding workflows, supporting ecosystems including DTV, Globo, FIFA, Roku, BetPlay and Blackhawk Network.
- Built Rust workers for high-volume event and insight pipelines, integrated AWS Bedrock and RAG/LLM decision support, and designed BPMN/Camunda/Temporal orchestration.
- Led Java-to-Go modernization using CPU, memory and latency profiles; redesigned asynchronous processing, private AWS networking and Aurora/AppSync/Valkey data paths for more efficient execution.
- Established API, schema and release standards; led architecture reviews, senior-engineer mentorship, production RCA and PCI/SOC2-aligned delivery with product, SRE and compliance stakeholders.

PROFESSIONAL EXPERIENCE

Fermyon Technologies India Private Limited (LINK, formerly ShoppinPal)

Senior Software Engineer / Engineering Mentor | May 2023 - Jan 2024

Environment: Python/FastAPI, Go, Java, Node.js, AWS, REST integrations, CI/CD, observability

- Designed backend systems and internal tools for data migration, partner integrations, workflow automation, QA traceability and operational dashboards.
- Delivered REST integrations for GoDaddy, Shopify, Clover, Square, Promenade and LightSpeed, preserving data integrity and backward compatibility across partner contracts.
- Modernized selected Java and Node.js components to Go to reduce runtime overhead and improve concurrency, CPU and memory behavior in high-volume integration jobs.
- Strengthened logging, QA automation, release controls and RCA-led fixes; mentored engineers through design reviews, debugging and production-readiness assessments.

Maverick Quantum India Private Limited (mavQ)

Software Development Lead | Aug 2022 - May 2023

Environment: Python/FastAPI, Java, Node.js, AWS Connect, API-first architecture, workflow automation, CI/CD

- Led architecture and delivery of no-code automation capabilities, with API-first services, reusable abstractions and workflow execution designed for scale.
- Owned service design, data models and test strategy through release and production support; developed shared database and platform components to shorten service rollout and improve consistency.
- Delivered AWS Connect customer-experience capabilities and contributed to Intelligent Document Processing automation in collaboration with product, operations and support teams.
- Mentored engineers on API design, testing and production readiness while coordinating delivery decisions across QA, DevOps and business stakeholders.

GoSprint Logistics Private Limited (Swift)

Platform Lead | Apr 2021 - Aug 2022

Environment: Rust, Python, ML processing, Go, Java, Node.js, AWS, MySQL, MongoDB, checkout, payments and logistics

- Led architecture of commerce and logistics capabilities spanning checkout, payments, courier orchestration, merchant workflows and partner integrations.
- Implemented Rust processing paths for high-volume shipment data feeding ML models, improving throughput and control over logistics datasets.
- Created SenseEngine for ML-driven shipment intelligence across 300K+ monthly orders, reducing cost per shipment through optimization logic.
- Defined resilient Java/Go service boundaries and backward-compatible APIs across payment, courier and partner systems; built BYOC and BYOP platforms for configurable orchestration.
- Worked with product and operations teams on release quality, incident response and production scalability, retaining hands-on ownership of implementation and diagnosis.

PROFESSIONAL EXPERIENCE

D-SYS Data Solutions Private Limited | Gwalior, MP

Employment and outsourced assignments | Dec 2014 - Nov 2020

D-SYS outsourced assignments on client payrolls: Omega Healthcare Management Solutions (Nov 2019); Collasys Global Services (Dec 2019 - Feb 2020); Technosoft Consulting (Feb 2020 - May 2020); Cooey Technologies (Jul 2020 - Nov 2020).

Senior Full-Stack Developer | Mar 2018 - Nov 2020

Environment: Python, Data Analysis, SQL, .NET MVC, C#, Entity Framework, Xamarin, Web APIs, Transportation / E-Way Billing Systems, Compliance Workflows

- Delivered enterprise full-stack systems with .NET MVC, C#, SQL, and Xamarin, including transportation and e-way billing workflows for national road-traffic compliance use cases.
- Expanded Python use into data analysis, structured data preparation, reporting, and model experimentation for analytics-heavy product work.
- Built backend foundations for safety/security product workflows using Node.js and Express.js, guiding Angular, Node.js, Android, and backend engineers.
- Extended Python, Go, and Rust-backed engineering into ML data collection, exploratory analysis, semi-supervised learning concepts, sentiment signals, recommendation logic, and multi-model experiments.
- Contributed to smart security systems with ML data collection and security-domain intelligence to improve operational performance.
- Owned backend and application features from design through implementation, debugging, deployment support, and client-facing resolution.

Full-Stack / Mobile Developer | Dec 2014 - Feb 2018

Environment: Java, Java OOP, Python/Django, .NET MVC, Entity Framework, Android, iOS, MySQL, MongoDB, AWS, Linux, P2P Server Management

- Delivered mobile, backend, marketplace, database, and production-server capabilities for client-facing product and enterprise engagements.
- Engineered Java-led client systems with Python/Django support for admin workflows, APIs, client portals, and database-backed applications.
- Implemented Android/iOS applications, server-side components, P2P server support, .NET MVC marketplace features, database designs, and operational deployment workflows.
- Supported D-SYS Data Solutions product and client-delivery initiatives across mobile, marketplace, database-backed workflow, and server operations work.
- Coordinated with distributed designers, developers, marketing teams, and business stakeholders to move requirements from concept to working production applications.

Aadi Infotech Pvt. Ltd. | Gwalior, MP

Software Developer / Software Intern | Feb 2010 - Feb 2014

- Progressed from Software Intern (2010-2012) to Software Developer (2012-2014), delivering transportation, education, social/image-editing, mobile and enterprise applications.
- Built Java-backed systems and Python/Django APIs and admin tools; helped redesign standalone applications into object-oriented, database-backed thin-client architectures.

Concurrent Freelance and Consulting | Dec 2014 - Nov 2019

Millennial Group Inc., Cloud Digital Arts, Cargo 3 IVS / FindACargo and independent clients: mobile apps, marketplace features, database workflows, server operations and release coordination.

EDUCATION

M.Tech in Artificial Intelligence and Machine Learning: BITS Pilani | Oct 2026 - Oct 2028 (expected); in progress.

Postgraduate Program in Artificial Intelligence and Machine Learning: The University of Texas at Austin; facilitated by Great Learning.

B.E. in Information Technology: Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV) | 2007 - 2012.

Ongoing study and planned practice: Agentic AI and tooling: tool calling, Model Context Protocol (MCP), LangChain, LangGraph, agent state, approvals and recovery; LLM evaluation, MLOps/LLMOps and inference optimization. Separate systems track: Linux systems programming, storage, Lustre, HPC and ultra-low-latency design.